

School of Information Technologies
Faculty of Engineering & IT

ASSIGNMENT/PROJECT COVERSHEET - GROUP ASSESSMENT

Unit of Study: SOFT2412

Assignment name: **Group project Assignment 2 - Scrum Software Development**

Tutorial time: Every Wednesday from 16:00 to 18:00

Tutor name: **Ronald Noronha**

DECLARATION

We the undersigned declare that we have read and understood the *University of Sydney Student Plagiarism: Coursework Policy and Procedure*, and except where specifically acknowledged, the work contained in this assignment/project is our own work, and has not been copied from other sources or been previously submitted for award or assessment.

We understand that failure to comply with the *Student Plagiarism: Coursework Policy and Procedure* can lead to severe penalties as outlined under Chapter 8 of the *University of Sydney By-Law 1999* (as amended). These penalties may be imposed in cases where any significant portion of my submitted work has been copied without proper acknowledgement from other sources, including published works, the internet, existing programs, the work of other students, or work previously submitted for other awards or assessments.

We realise that we may be asked to identify those portions of the work contributed by each of us and required to demonstrate our individual knowledge of the relevant material by answering oral questions or by undertaking supplementary work, either written or in the laboratory, in order to arrive at the final assessment mark.

Project team members				
Student name	Student ID	Participated	Agree to share	Signature
1. Adam Behrmann	490612282	Yes	Yes	Adam
2. Lehan(Hans) Yang	500136245	Yes	Yes	Lehan Yang
3. Alex Tan	490379282	Yes	Yes	Alex
4. Shimiao Sun	480142841	Yes	Yes	Shimiao
5. Sung Bin Park	500574470	Yes	Yes	Sung Bin

Sprint 1 Report

Spring planning Wednesday 13/10 4pm

Current sprint backlog

BT Sprint 1 13 Oct – 20 Oct (11 issues)		94	0	0	Complete sprint	...
BT-6	As a customer I want to be able to choose movies to book tickets for and select options, so that I can conveniently book a movie.	13	↑	TO DO	SP	
BT-29	As a customer I want to be able to choose which cinema to watch the movie at so that I can watch the movie at a convenient location.	5	↑	TO DO	HL	
BT-7	As a customer I want to be able to cancel a booking transaction so that I can change my mind.	3	↑	TO DO	AP	
BT-9	As a customer, I would like to filter movies by classification, to find age appropriate movies for my children	13	↑	TO DO	AP	
BT-4	As a guest user/customer I want to filter for selected cinemas and cinema status (gold/silver/bronze) so that I can better select a cinema.	5	↑	TO DO	AP	
BT-10	As a customer, I would like to filter movies by price, so I can see which movies are in my budget	5	↑	TO DO	AP	
BT-18	As a cinema staff, I want to know how many seats have been booked and are available for each movie session, so that I can stop selling tickets when one session is full.	8	↑	TO DO	SS	
BT-11	As a cinema staff, I want to modify movie data file, so that I can update the newest movie information of our cinema to let customers know.	13	↑	TO DO	SS	
BT-5	As a guest I should be prompted to login/register when attempting to book a movie so that I am able to book the movie.	3	↑	TO DO	AP	
BT-30	As a guest I want to be able to register an account, so that my bookings are saved.	13	↑	TO DO	SP	
BT-31	As a guest I want to be able to login to my account, so that I can view my bookings.	13	↑	TO DO	AP	
+ Create issue						

Product backlog

Backlog (12 issues)		56	0	0	Create sprint
BT-1	As a guest/customer I want to be able to view upcoming movies and times so that I can select a movie to watch	8	=	TO DO	
BT-8	As a customer I want my booking transaction to be automatically cancelled after 2 minute timeout so that my security uncompromised.	3	↓	TO DO	
BT-12	As a cinema staff, I want to gain the information about upcoming movies and show with their detailed data, so that I can know what I should do while updating movie data for o...	3	=	TO DO	
BT-13	As a user, I would like to have an option to remember my log in details, so I dont have to manually enter my details every time		↓	TO DO	
BT-14	As a customer, I would like to have a movie ranked on number of bookings, so I could find out popular movies	4	↓	TO DO	
BT-15	As a cinema manager, I want to add/remove cinema staff members, so I can control the employees of the cinema on the system.	8	↓	TO DO	
BT-16	As a cinema manager, I want to have a summary of cancelled transactions by the user as a file, so that I can see if there are any problems that are causing the cancellations.	8	↓	TO DO	
BT-19	As a customer, I would like to get a notice if there are cheaper session option for the film I chose	3	=	TO DO	
BT-20	As a customer, I would like to get a option to choose to rent or buy a 3d glasses if I chose to watch a 3d movie	3	=	TO DO	
BT-21	As a customer, when the credit card declined, I'd like to receive a message to know what the type of error I get, like insufficient funds or card locked	8	=	TO DO	
BT-22	As a cinema manager, I want to have a button that I can cancel the specific session if cinema faced some issues, and automatic refund to customers.	8	=	TO DO	
BT-28	As a user, I would like to book a ticket using a gift card that a friend has bought me		=	TO DO	
+ Create issue					

Standup Friday 15/10 9am

Attendees: all

Meeting minutes

Adam :

- no progress yet
- features to be done sunday

Alex :

- worked on db tables
- submitted PR
- will work on integrating with java

Hans:

- No progress yet
- Will complete features by tomorrow evening

Sung Bin:

- coming up with login system
- to discuss with Adam about BT31

Shimiao:

- worked on example db connection in java
- will complete modification functions for db by tomorrow

Standup Sunday 17/10 9am

Attendees: all

Meeting minutes

Adam :

- created Account class
- contains movie bookings
- worked on login
- will implement logic for booking
- To create a list of user commands for user to perform actions e.g. booking, logging in and registering

Alex :

- reviewed PR 1
- submitted PR 2 to be reviewed
- to work on integrating db with java

Hans:

- Reviewed PR 2
- Create Movie/Movies/Cinemas/Cinema/Booking class

Sung Bin:

- created system class which has a login method

- todo: review adam's PR

Shimiao:

- prepared statement confusion
- worked on data modification functions
- Will improve sql searching and do seats checking functions

Standup Tuesday 19/10 3:30pm

Attendees: All but Sung-bin

Adam

- how to interact db with sql
- discuss UML?
 - to avoid too many merge conflicts
 - Adam to create UML? yes/no?
 - commandline system?
 - google doc;
- blocked by not sure how to interact with SQL db from System.java

Alex

- wrote UI code
- created more db tables

Shimiao

- implemented functions for db

Hans

- stuck on db
- will implement booking operation

Sprint review

Feedback from client:

- reconsider flow of the UI stages
 - e.g. consider having a cancel point at each stage to take you back to the main stage
- need to have options to select further division for seats (front, middle, back)
- for next week, suggested to finish off customer completely

Incomplete issues

[View in issue navigator](#)

Key	Summary	Issue type	Epic	Status	Assignee	Story points
BT-5	As a customer I should be prompted to login/regist...	Story		IN PROGRESS	AB	3
BT-7	As a customer I want to be able to cancel a transacti...	Story		IN PROGRESS	AB	3
BT-31	As a customer I want to be able to login to my accou...	Story		IN PROGRESS	AB	13

Completed issues

[View in issue navigator](#)

Key	Summary	Issue type	Epic	Status	Assignee	Story points
BT-6	As a customer I want to be able to choose movies to...	Story		DONE	HL	13
BT-9	As a customer, I would like to filter movies by classi...	Story		DONE	A	13
BT-10	As a customer, I would like to filter movies by price, ...	Story		DONE	A	5
BT-11	As a cinema staff, I want to modify movie data file, s...	Story		DONE	SS	13
BT-18	As a cinema staff, I want to know how many seats ha...	Story		DONE	SS	8
BT-29	As a customer I want to be able to choose which cin...	Story		DONE	HL	5
BT-4	As a guest user/customer I want to filter for selecte...	Story		DONE	A	5
BT-30	As a guest I want to be able to register an account, s...	Story		DONE	SP	13

Sprint retro Thursday 21/10 9:30pm

Adam

What went well?

Consistent communication with Sung Bin and myself allowed us to create the logic for most user operations in our own branches.

What didn't go well?

I was unable to integrate my java code with the sql database in time for the demo, which resulted in a demo with less functionality than I would have liked.

What can be improved and how?

To resolve this, I will still be writing java code, but another teammate, with better sql knowledge than myself, will integrate the java code to the sql database.

Alex

- **what went well?:** easy to communicate to team members, quick response
- **what didn't go well? how to improve:** PRs not being reviewed sufficiently fast to ensure progress; can be improved by notifying team each time there is a PR

Shimiao

- **What didn't go well?** Waste time using an inefficient way for the sql query at first.
- **What went well?** Quick response and finish tasks as early as possible.

- **How to improve?** Communicate with the team before having some thoughts on time to make sure that the idea is correct and efficient.

Hans

- **What went well?:**
- We are successfully using branches to manage the repo in this project.
- Successfully build Jenkins and PostgreSQL, have a better performance than previous project.
- **What didn't go well? how to improve:**
- Not familiar with using SQL in java code, faced some bugs.
- Git merge seems always have conflict issues, but have not been solved very well.
- Next sprint I need to finish my ticket earlier to make sure we have more time to do integration and communications.

Sung Bin

- **What went well?**
Easy communication with the team, consistent branching for different versions, quick feedback.
- **What didn't go well? How to improve?**
I was unfamiliar with using java integration with the sql database, resulting in 2 different versions of the main app/UI. Starting on the branch that has successfully set connection with sql, I believe it's able to implement the same functionality/logic on the new branch.

CI/CD

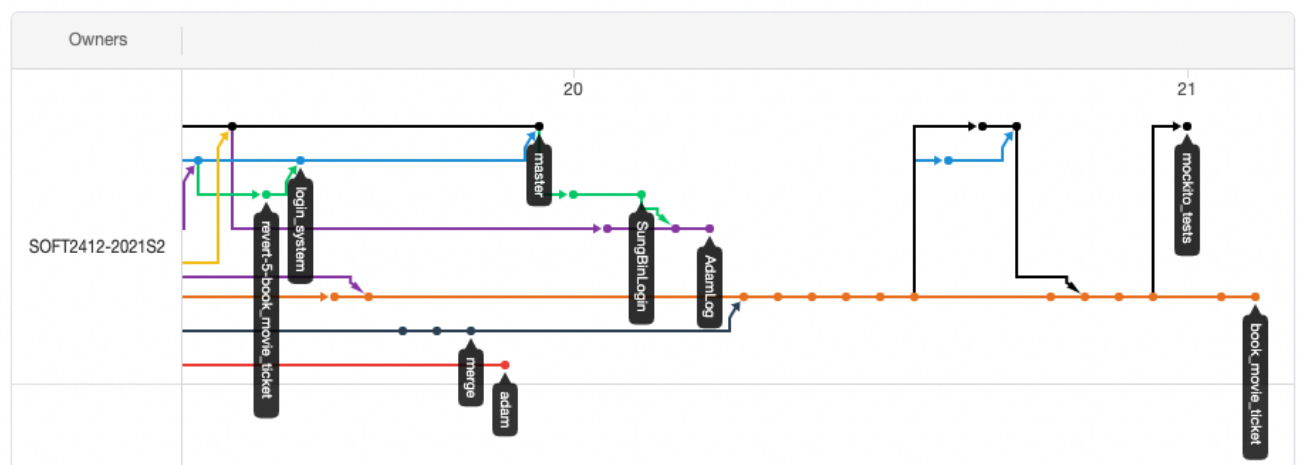
We have set up and using GitHub, Gradle, Junit and Jenkins during the development.

For GitHub, we have divided tasks into subtasks as branches. After every branch is developed, we are using the pull request function on GitHub to integrate branches together. In the pull request, we faced and solved conflicts between two branches built by different people. And also for each pull request, we need two approval before we can merge it to the master branch, in order to make sure the code quality.

Overview	Yours	Active	Stale	All branches	Search branches...	
All branches						
 master	Updated 2 days ago by spar0603		Default	Change default branch		
mockito_tests	Updated 1 hour ago by Alex		5 22	#13	Open	
book_movie_ticket	Updated 23 hours ago by Alex		5 21	#8	Open	
AdamLog	Updated 2 days ago by behrmdv		0 5	#11	Open	
SungBinLogin	Updated 2 days ago by root		0 2	#10	Merged	
adam	Updated 2 days ago by root		16 2	#4	Open	
merge	Updated 2 days ago by Alex		5 6	#9	Closed	
login_system	Updated 2 days ago by lyan3310		9 0	#3	Merged	
revert-5-book_movie_ticket	Updated 2 days ago by lyan3310		10 0	#7	Merged	
shimiao	Updated 3 days ago by =		9 0	#6	Merged	
movie_filtering	Updated 7 days ago by Alex		16 1	#1	Open	

Network graph

Timeline of the most recent commits to this repository and its network ordered by most recently pushed to.



Gradle as a build automation tool, we set it to build our code and run the CLI for users to use. Also we have added the jacoco dependency to generate the coverage report after the test ran.

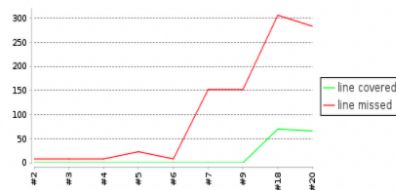
We also used Junit to do the test of our code.

Jenkins was installed in a docker image and hosted in Hans' home server, which can access via <http://home.xsource.cc:8082/>, thanks to the memory and cores of cpu increase, this Jenkins server can finish build operation in a much shorter time than the previous one in ATM project. Also it runs more stable and has not faced a crash yet. As same in ATM project, we use jenkins to do an automatic build and test on each commit to GitHub, also we use jenkins to automatically annotate the status of a

commit on GitHub. In the Jenkins code coverage report generation, we set to exclude calculate the ui's code in, while also set generate report even if the test failed.

JaCoCo Coverage Report

[Download jacoco.exec binary coverage file](#)



Overall Coverage Summary

name	instruction	branch	complexity	line	method	class
all classes	21% M: 848 C: 229	19% M: 26 C: 6	27% M: 55 C: 20	19% M: 283 C: 66	28% M: 41 C: 16	42% M: 7 C: 5

Coverage Breakdown by Package

name	instruction	branch	complexity	line	method	class
Assignment2_SOFT2412	M: 434 C: 172 28%	M: 20 C: 6 23%	M: 36 C: 17 32%	M: 122 C: 46 27%	M: 25 C: 13 34%	M: 7 C: 4 36%
Assignment2_SOFT2412.data	M: 414 C: 57 12%	M: 6 C: 0 0%	M: 19 C: 3 14%	M: 161 C: 20 11%	M: 16 C: 3 16%	M: 0 C: 1 100%

Individual contributions

Github repo https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412

- list all the Jira tickets you've been assigned and working on
- for the completed tickets, link the PR/commit/branch that you worked on that for that ticket

Alex

- Jira tickets BT4, 9, 10
- All three tickets completed in PR 9
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/9/commits/bbbd2109098b43aa1f6574c5b9deb3d31c57e528

Shimiao

- Jira tickets BT11, BT18
- These two tickets completed in PR 2
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/2

Hans:

- Set up gradle, Jenkins server and SQL server.
- Jira tickets BT 6(23, 24) and BT 29
- All these tickets completed in branch book_movie_ticket:
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/tree/book_movie_ticket

Adam:

- Jira tickets BT 5, BT 7 and BT 31

- Most of the logic for these tickets was implemented in PR 11, however there is still some logic remaining to be done, and the code has yet to be integrated with the SQL database, which is why the tickets still show as 'In Progress.'
- Link to PR
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/11/files

Sung Bin:

- Jira tickets BT 30
- Implemented the main methods/ class structure for the system.
- Ticket completed in PR 3, PR 10
- The branches were "login_system", merged to master as a basic class structure, and improved in the "SungBinLogin" branch.
- After finishing the login system, made a basic command line input system that can navigate the user through the system, according to the sample scenario above. This system takes commands such as LOGIN, REGISTER, BOOK, CINEMAS, MOVIES, QUIT, CANCEL.
- Link to PR
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/3/files
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/10/files